SINGAPORE DIMENSIONS • GRADE 5A • CHAPTER 2

Fractions: Adding with Unlike Denominators

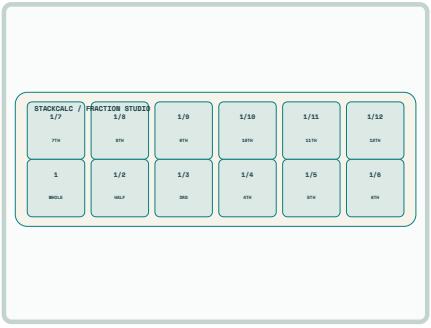
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.5.NF.A.1

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Fraction Tray

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Place 1/2 tower (42mm) and 1/3 tower (28mm). Test 1/6 blocks (14mm) to match height (70mm = 5×14mm).

Sketch Model on Paper:
Draw two rectangle bars partitioned into 6ths.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE: 123 Base-10 Arithmetic

1.	.	Step 1	Execute [.]
2.	1	Step 2	Execute [1]
3.	.	Step 3	Execute [.]
4.	2	Step 4	Execute [2]

EXPECTED LCD X: 3/4 (height 63.0mm)

1 **Add 1/3 + 1/6.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **Add 2/3 + 1/4.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **Subtract 3/4 - 1/2.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **Add 3/5 + 1/10.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Chapter 2

Fractions: Adding with Unlike Denominators • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Fraction Studio Towers (84.0mm Whole)

CAD Blueprint: fraction_tray

Target Device: SC-6 Elementary

CAD / STL Source File:

Education/scad/fraction_studio_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Teal 3x1 Bin & 1/1 to 1/12 Color-Coded Towers (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Add 1/3 + 1/6.** Standard: CCSS.MATH.5.NF.A.1	1/2	[.] [1] [.] [3] [ENTER] [.] [1] [.] [6
2	**Add 2/3 + 1/4.** Standard: CCSS.MATH.5.NF.A.1	11/12	[.] [2] [.] [3] [ENTER] [.] [1] [.] [4
3	**Subtract 3/4 - 1/2.** Standard: CCSS.MATH.5.NF.A.1	1/4	[.] [3] [.] [4] [ENTER] [.] [1] [.] [2
4	**Add 3/5 + 1/10.** Standard: CCSS.MATH.5.NF.A.1	7/10	[.] [3] [.] [5] [ENTER] [.] [1] [.] [1

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Common error: adding numerators and denominators (1/2 + 1/3 ≠ 2/5). Physical height disproves this instantly.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.